

Liraglutide with Lifestyle Intervention in Adolescents with Overweight/Obesity, Nonalcoholic Fatty Liver Disease, and Type II Diabetes Mellitus.

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There is a strong interplay between nonalcoholic fatty liver disease (NAFLD), obesity, insulin resistance, and type II diabetes mellitus (T2DM). Liraglutide, a glucagon-like-peptide-1 (GLP-1) analogue, is FDA approved for T2DM in children 10 years or older and more recently approved for chronic weight management in children 12 years or older with obesity. GLP-1 analogues have also been shown to reduce liver enzymes and improve liver histology. We report two adolescent females with T2DM and biopsy proven nonalcoholic steatohepatitis (NASH) refractory to lifestyle intervention who were safely treated with liraglutide with associated weight loss and liver enzyme improvement. This is the first case series reporting use of liraglutide in pediatric NASH. Liraglutide should be considered in pediatric patients with overweight/obesity, NAFLD, and T2DM.

Risk of Suicidal Ideation or Attempts in Adolescents With Obesity Treated With GLP1 Receptor Agonists.

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Glucagon-like peptide 1 receptor agonists (GLP1R) are increasingly being used for the treatment of obesity in adolescents. It is currently unknown whether GLP1R treatment is associated with suicidal ideation or attempts in this population. To investigate the association between GLP1R initiation and suicidal ideation or attempts in adolescents with obesity. Retrospective propensity score-matched cohort study using electronic health records from the TriNetX global federated network between December 2019 and June 2024. The analysis included data from 120 health care organizations, mainly from the USA. Participants were adolescents aged 12 to 18 years with a diagnosis of obesity and evidence of an antiobesity GLP1R prescription or lifestyle intervention without GLP1R within the following year. Cohorts were balanced for baseline demographic characteristics, psychiatric medications and comorbidities, and diagnoses associated with socioeconomic status and health care access using propensity score matching. Initial prescription of GLP1R (study cohort) or lifestyle intervention without GLP1R (control cohort). Incidence of suicidal ideation or attempts based on International Statistical Classification of Diseases and Related Health Problems, Tenth Revision codes recorded in patient electronic health records during 12 months of follow-up. Diagnoses of upper respiratory tract infections (URTI) were used as negative control outcomes, and gastrointestinal symptoms (GI) were used as positive control outcomes. A total of 4052 adolescents with obesity and a concomitant antiobesity intervention were identified for the GLP1R cohort and 50 112 were identified for the control cohort. Propensity score matching resulted in 3456 participants in each balanced cohort. Prescription of GLP1R was associated with a 33% reduced risk for suicidal ideation or attempts over 12 months of follow-up (1.45% vs 2.26%; hazard ratio [HR], 0.67; 95% CI, 0.47-0.95; $P = .02$) and a higher rate of GI symptoms (6.9% vs 5.4%; HR, 1.41; 95% CI, 1.12-1.78; $P = .003$) but no difference in rates of URTI diagnoses. In this study, adolescents with obesity prescribed a GLP1R had a lower incidence of suicidal ideation or attempts compared with matched patients not prescribed GLP1R who were treated with lifestyle intervention. These results suggest a favorable psychiatric safety profile of GLP1R in adolescents. The detected reduction in HRs for suicidal ideation among adolescents with obesity prescribed GLP1R suggests potential avenues for future research.

Adolescents with obesity treated with exenatide maintain endogenous GLP-1, reduce DPP-4, and improve glycemic control.

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GLP-1 receptor agonists (GLP-1RA) are increasingly used to treat adolescent obesity. However, the effect on endogenous GLP-1 secretory patterns following treatment in adolescents is unknown. The GLP-1RA exenatide was shown to significantly lower BMI and 2-hour glucose in adolescents with obesity, in the placebo-controlled, randomized controlled trial Combat-JUDO. The aim of this study was to evaluate effects of weekly injections of 2 mg exenatide extended release on secretory patterns of endogenous hormones during OGTT. This study was a pre-planned sub-study of the Combat-JUDO trial, set at the Pediatric clinic at Uppsala University Hospital, Sweden and Paracelsus Medical University, Austria. 44 adolescents with obesity were included and randomized 1:1 to treatment:placebo. 19 patients in the treatment group and 18 in the placebo group completed the trial. Before and after treatment, GLP-1, glucose, insulin, glucagon and glicentin levels were measured during OGTT; DPP-4 and proinsulin were measured at fasting. A per-protocol approach was used in the analyses. Exenatide treatment did not affect GLP-1 levels during OGTT. Treatment significantly lowered DPP-4, proinsulin and the proinsulin-to-insulin ratio at fasting, increased glicentin levels but did not affect insulin, C-peptide or glucagon levels during OGTT. Weekly s.c. injections with 2 mg of exenatide maintains endogenous total GLP-1 levels and lowers circulating DPP-4 levels. This adds an argument in favor of using exenatide in the treatment of pediatric obesity. [clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT02794402), identifier NCT02794402.

Acute Kidney and Liver Injury Associated With Low-Dose Liraglutide in an Obese Adolescent Patient.

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In 2020, the US Food and Drug Administration approved liraglutide (glucagon-like-peptide-1-receptor-agonist) as an adjunctive therapy for weight management in adolescents aged 12 to 18 years in combination with a reduced-calorie diet and increased physical activity. The 2023 American Academy of Pediatrics guidelines recommend pharmacotherapy with glucagon-like-peptide-1-receptor-agonist as a second-line therapy in obesity management. Although reports in adults have suggested a link between liraglutide and adverse effects including hepatic injury and acute kidney injury (AKI), these effects have not previously been reported among adolescents treated with liraglutide for weight loss. We present a 17-year-old male who developed AKI and evidence of hepatic injury (significant elevation of hepatic transaminases) after 3 months administration of the lowest dosage of liraglutide (0.6 mg/day) for management of class III obesity. The patient experienced significant loss of appetite, weight loss, and melancholy during the treatment period. One month after discontinuing liraglutide, his mood had improved, his liver enzymes had returned to normal, and AKI had resolved. The Adverse Drug Reaction Probability Scale suggested a high likelihood of a causative association between liraglutide and his symptoms. Our report highlights the importance of vigilance in monitoring for these potential adverse effects among adolescents treated for obesity with any dose of liraglutide.

Impact of anti-tumor necrosis factor alpha antibodies administered to pregnant women

with inflammatory bowel disease on long-term outcome of exposed children.

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Prenatal exposure to anti-tumor necrosis factor α (TNF- α) antibodies seems to be safe for fetal development. Data on long-term outcome of exposed children are missing. Our aim was to assess long-term postnatal development of children exposed to anti-TNF- α during pregnancy. Consecutive children aged ≥ 12 months exposed to anti-TNFs prenatally for maternal inflammatory bowel disease in 3 centers in the Czech Republic were enrolled. Data on psychomotor development, infections, antibiotics, vaccination, and allergy were retrospectively obtained from mothers, treating pediatricians, and children's vaccination cards. Furthermore, standardized laboratory tests on humoral and cellular immunity were performed. Twenty-five children exposed to biologicals were included (median age, 34 mo; range, 14-70 mo). All children had normal growth, and all but 1 had normal psychomotor development. Majority (80%) experienced at least 1 infection (mainly respiratory), and 60% of infants received antibiotics, 32% of those within the first year of life. Vaccination was undertaken according to vaccination protocol to 23 infants (92%). Fifteen children also had tuberculosis vaccination without serious complication. Immunological investigation was performed with 17 children (68%). Cellular immunity was normal in all infants, and 7 children had mild decrease in IgA and/or IgG immunoglobulins without clinical significance. All children had a detectable serologic response to vaccination. Exposure to anti-TNF- α antibodies seems to be safe for growth and psychomotor development of children, although clinical significance of relatively high frequency of infections and antibiotic use among infants remains questionable because of the lack of a control group. Continuous follow-up of exposed children is absolutely warranted.

Exposure to biologic therapy before and during pregnancy in patients with psoriasis: Systematic review and meta-analysis.

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Biologicals have transformed the management of severe disease phenotypes in psoriasis and are often prescribed in women of childbearing age. However, information on safety of biologicals in pregnancy are lacking. We conducted a systematic review and meta-analysis aimed to describe the characteristics and pregnancy outcomes in women with psoriasis exposed to biologics within 3 months before or during pregnancy, and to estimate the pooled prevalence of spontaneous, elective and total abortions, and congenital malformations in their newborns. Bibliographic searches were performed in the PubMed, Embase, Scopus and Web of Science databases up to 14 April 2022. No restrictions on sample size or publication date were applied. Review performance complied with PRISMA guidelines, and two reviewers assessed randomized controlled trials and nonrandomized studies reporting pregnancy outcomes in women exposed to biologics indicated for psoriasis during the pre-gestational and/or gestational period. Studies focusing on rheumatologic or gastroenterological immune-mediated inflammatory diseases were excluded. Regardless of data heterogeneity, a random-effects model was used to pool prevalence estimates. We included 51 observational studies, involving 739 pregnancies exposed to approved biologics for psoriasis. Administration was mostly (70.4%) limited to the first trimester, and the most common drug was ustekinumab (36.0%). The estimated prevalence of miscarriage was 15.3% (95% confidence interval [CI] 12.7-18.0) and elective abortions, 10.8% (95% CI 7.7-14.3). Congenital malformations occurred in about 3.0% (95% CI 1.6-4.8) of live births exposed to biologics during pregnancy. Altogether, exposure to biologics for psoriasis during pregnancy and/or conception does not seem to be associated with an increased risk of miscarriage/abortion or congenital malformations,

showing similar rates to the general population. These results suggest that biologic drugs are safe and pose an acceptable risk to the fetuses/neonates.

Safety of biologic immunosuppressants in pregnant women with immune-mediated inflammatory diseases.

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Immune-mediated inflammatory diseases (IMIDs) typically affect women of childbearing age. One of the challenges in treating these women during pregnancy is to manage the disease while minimizing or avoiding the use of disease-modifying antirheumatic drugs (DMARDs) that may increase the risk to the mother or fetus. Biologic therapy has transformed the management of these patients. This study aimed to evaluate the maternal-fetal safety and perinatal outcomes in pregnant women with IMID exposed to biologic DMARDs either preconceptionally or during pregnancy and compare them with women using conventional DMARDs and a group of healthy pregnant women. We conducted a retrospective study with prospective follow-up of pregnant women with IMID at a single center. We analyzed baseline maternal demographic characteristics, diseases, DMARDs, and maternal-fetal outcomes. A cohort of 244 pregnancies was studied. One hundred twenty-eight patients met classificatory criteria for rheumatic and musculoskeletal diseases (RMD) or inflammatory bowel disease (IBD), and 116 pregnancies of healthy women were evaluated from the same study period. One hundred and one pregnancies in IMID patients (89.84 %) occurred under immunosuppressive treatment, 78.91 % of IMID pregnancies were under cDMARD (33.59 % exclusive cDMARD), 56.25 % under bDMARD, and 27.34 % under oral glucocorticoids. Anti-TNF was the most frequent (88.88 %) bDMARD and was used in 50.78 % of the IMIDs. There was at least one flare in 37.10 % of the IMID pregnancies, and 9.38 % experienced more than one. Among flares, 43.48 % happened in the first trimester, 34.78 % in the second trimester, and 19.57 % in the third. Flares were more frequent in the RMD patients compared with IBD ($p = 0.041$; OR 2.15, 95%CI: 1.03-4.52). Flare was associated with discontinuation of bDMARD before the eighth week of gestation ($p = 0.016$), but especially in the second ($p = 0.042$) and third trimester ($p = 0.012$). Maternal infections were an infrequent complication overall (7.66 %), although more frequent in patients with IMIDs ($p = 0.004$) but were not associated with cDMARD or bDMARD. IMID patients needed assisted reproductive techniques (ART) more often ($p = 0.001$, OR 2.83, 95%CI: 1.02-7.90). More cesarean sections were performed in gestations under treatment with bDMARD ($p = 0.020$) and especially in those under treatment with anti-TNF. Aneuploidies calculation risk and fetal malformations were not correlated with DMARDs (cDMARDs, bDMARDs, or its combination) nor with any of the DMARDs individually preconceptionally or during gestation. Small for gestational age (SGA) newborns were higher in patients with IMIDs however, it was not associated with DMARD use. In general, patients with IMIDs who require treatment with bDMARDs have a more severe or refractory disease prior to gestation. In our cohort, we found a higher risk of flare among patients with bDMARDs, especially when those were suspended early. Among maternal outcomes, we found that IMID patients needed ART more often. This is probably, first of all, because of maternal age. Among fetal outcomes, there are no differences in congenital malformations in the IMIDs and healthy patients and were not correlated with DMARDs. The use of bDMARDs was effective in disease control and safe from a maternal-fetal point of view, with no increase in prematurity, SGA, malformations, or infections.

Are biological drugs safe in pregnancy?

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The introduction of biological therapies has significantly improved the outcome of inflammatory rheumatic diseases. As most of these diseases affect women and men in childbearing age, some concerns have been voiced as to the safety of these drugs in relation to reproduction and pregnancy. Data from many hundreds of pregnancies in patients affected by inflammatory bowel disease and inflammatory arthritis have suggested that exposure to anti-TNF therapies at conception and/or during pregnancy is not associated with adverse pregnancy outcomes or any increase in congenital abnormalities. However, the exposure to anti-TNF α agents, particularly to monoclonal antibodies, in late pregnancy is associated with high drug levels in the newborn and their long-term effects on children remain unknown. Therefore, limiting the use of anti-TNF α to the first 30 weeks of pregnancy is recommended to reduce fetal exposure. Live-virus vaccines should be given only when levels of anti-TNF α drugs are undetectable in the serum of infants. Studies suggest that many of these drugs do enter breast milk in small amounts, but the extent to which the infant absorbs them is less clear. Limited reports have not suggested adverse pregnancy outcomes in women whose partners were exposed to anti-TNF therapies at the time of conception. Pregnancy data for rituximab, abatacept, anakinra, tocilizumab and belimumab are limited and their use in pregnancy cannot currently be recommended.

Safety issues of biologics in pregnant patients with rheumatic diseases.

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The safety of biologic agents during pregnancy is still being investigated. Tumor necrosis factor α (TNF- α) inhibitors are the only well-studied biological drugs in pregnancy; they do not appear to be teratogenic but increase the risk of infection after birth when given in late pregnancy. The long-term effects in exposed children are, at present, unknown. An increased risk of infection is a concern for all biologics and the risk increases further should combination with glucocorticoids be necessary. Experiences with rituximab, abatacept, anakinra, tocilizumab, and belimumab in pregnancy are limited. These drugs should be avoided during pregnancy or used only when no other option is available for treatment of serious maternal disease.

The safety of drugs for inflammatory bowel disease during pregnancy and breastfeeding: the DUMBO registry study protocol of GETECCU.

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Active disease during conception and pregnancy in women with inflammatory bowel disease (IBD) increases the risk of pregnancy complications and adverse neonatal outcomes. The use of IBD treatments during pregnancy should be weighed against their adverse effects on the neonate, but longer-term safety data and data on serious infection rates and malignancies postnatally are lacking, particularly for newer drugs, such as tofacitinib, vedolizumab and ustekinumab. This ongoing, prospective registry study being conducted at 70 centres in Spain is enrolling pregnant women who are ≥ 18 years, are at any point in pregnancy up to the end of the second trimester and have a diagnosis of Crohn's disease, ulcerative colitis or unclassified IBD. Patients will receive treatment decided independently by their IBD specialist. Each incident gestation will be followed up through pregnancy and the first 4 years postnatally. Three cohorts will be compared: biologicals exposed, immunomodulatory exposed and non-exposed. The

primary endpoint is the risk of severe infection in newborns postnatally up to 4 years of age; other endpoints include serious adverse events (SAEs) such as pregnancy and delivery complications, neonatal SAEs, development [Ages and Stages Questionnaire-3 (ASQ3)], and malignancy incidence, up to 4 years of age. IBD specialists will collect maternal data (baseline/end of each trimester/1 month post-delivery), neonatal birth data, and the SAE and ASQ3 data in children exposed during pregnancy, reported every 3 months by the mother. Statistical analysis will include summary statistics for quantitative variables, comparisons of qualitative variables with significance set at $p < 0.025$ and a binary logistic regression model to determine the risk factors for severe infections. Enrolment began in September 2019 and study completion is expected in September 2028. This prospective, controlled study will provide evidence on the long-term safety profile in children after intrauterine and lactation exposure to biological and immunomodulatory IBD treatments, including data on postnatal severe infections, development and malignancies. NCT03894228.
